EXECUTIVE SUMMARY

Predicting Zambia’s 2026 FRA Maize Floor Price

Boldwin Mweemba • Data & Financial Analyst, Lusaka, Zambia • June 2026

Background & Objective

The Food Reserve Agency (FRA) sets a single maize floor price - in Kwacha per 50 kg bag - once a year, making it a critical anchor for Zambia’s food security and agricultural income. The announcement, expected in mid-2026, affects smallholder farmers, commodity traders, and the government’s fiscal position. Despite its importance, the price is rarely modelled in advance.

This analysis builds a data-driven model to forecast the 2026 FRA maize floor price ahead of the announcement, using macroeconomic, commodity, and agricultural data from 2002 to 2025.

Data & Sources

The model draws on 24 annual observations spanning 2002-2025, combining live API data with verified embedded values:

Feature	Source	Access
FRA floor price	FRA announcements / MSU-FSRP academic table	Embedded (no API)
Inflation (annual %)	World Bank API (FP.CPI.TOTL.ZG)	Live
Exchange rate (ZMW/US$)	World Bank API (PA.NUS.FCRF)	Live
Maize harvest (tonnes)	ZamStats Crop Forecast Survey	Embedded (verified)
Global maize price ($/t)	FRED API (PMAIZMTUSDA)	Live

All prices are expressed in today’s rebased kwacha (pre-2013 figures divided by 1,000). Inflation and exchange rate values for 2002-2024 are verified World Bank actuals; 2025-2026 are best-available estimates.

Methodology

A Ridge regression model (regularised linear regression, $\alpha = 1.0$) is trained on five features: the previous year’s FRA price, the kwacha/USD exchange rate, annual inflation, the maize harvest, and an election-year flag. Features are standardised before fitting so that coefficients are directly comparable. Model accuracy is evaluated using leave-one-out cross-validation (LOO-CV), which is appropriate for small annual samples as it uses every year as an unseen test case in turn.

Four feature sets were compared via LOO-CV. The base five-feature model achieved the lowest mean absolute error (MAE = K14.5 per 50 kg bag, $R^2 = 0.93$), with every additional feature making accuracy worse - a textbook signal of overfitting on a small dataset.

Key Findings

- **Price dynamics:** The price is sticky and almost entirely upward.
- The FRA price fell only twice in 24 years (2003 and 2017). Every other year it was held flat or raised, making last year’s price the strongest single predictor.
- **Dominant drivers:** The previous year’s price and the kwacha/USD exchange rate account for almost all of the model’s predictive power (standardised coefficients of +57 and +35 respectively).

- A weaker kwacha raises import-linked input costs, which the FRA has historically passed on through the floor price. All other factors - harvest, inflation, and the election-year flag -are minor by comparison.
- **Feature selection:** Adding fertiliser cost and global maize price raised the CV error, confirming they add redundancy rather than new information on this sample size.
- **Election-year effect:** 2026 is an election year, and the FRA has never cut the price in an election year in the historical record. However the election coefficient is negligible in the model; the pattern is consistent but the sample of election years is too small to be statistically meaningful.

2026 Prediction

Predicted 2026 FRA price	Indicative range (LOO-MAE)
K356 per 50 kg bag	**K341 – K371**

2025 baseline: K340 per 50 kg bag → predicted change: +K16 (+4.7%)

Exchange rate assumption: The prediction uses a 2026 exchange rate of ZMW 28 per US dollar, extrapolated from the World Bank's 2024 annual average of ZMW 26.17. The actual 2026 rate has since strengthened significantly (currently ZMW 17–19). Under current market conditions, the model projects approximately K310-K330. The interactive dashboard allows users to test predictions under any exchange rate scenario.

Caveats & Limitations

- **Small sample:** The model is based on approximately 24 annual data points. Treat the prediction as directional, not exact.
- **Political decision:** The FRA price is ultimately set by a committee with discretion. Presidential overrides have occurred; sudden cuts or outsized jumps cannot be modelled.
- **Data quality:** FRA price years 2010–2012 and 2014–2015, and the 2002–2016 maize harvest figures, are best-available estimates rather than fully verified actuals.
- **Exchange rate sensitivity:** The forecast is sensitive to the assumed 2026 exchange rate. A stronger kwacha would imply a lower predicted price.

Conclusion

The data points to a 2026 FRA maize floor price in the mid-K350s - a modest, single-digit-percentage increase on the 2025 price of K340, consistent with the FRA's pattern of steady upward adjustments. The prediction is driven almost entirely by two factors: price stickiness (last year's K340 anchors this year's decision) and the kwacha's long-run depreciation trajectory. The election-year flag, harvest, and inflation provide minor supporting signals.

The main uncertainty is the exchange rate. If the kwacha's 2026 strengthening is sustained, the economic case for a large increase weakens and the more likely outcome is a hold or a small rise near K340. The interactive dashboard (zambia-fra-maize-price.streamlit.app) allows any assumption to be tested in real time.